Name of Student: Minisha Singh	**PRN: 2507022292**
Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 16	
TITLE : User-Defined Exceptions	

Problem Statement

Write a Java program to create and handle a user-defined exception.

Learning Objectives

To create and implement custom exceptions for application-specific error handling.

Exercise

1. **Create a Voting System that throws a custom exception if age is below 18**

Code:

```
import java.util.Scanner;

class InvalidAgeException extends Exception {
    InvalidAgeException(String message) {
        super(message);
    }
}

public class Main {

    static void checkAge(int age) throws InvalidAgeException {
        if (age < 18) {
            throw new InvalidAgeException("You are not eligible to vote.");
        } else {
            System.out.println("You are eligible to vote.");
        }
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter your age: ");
        int age = sc.nextInt();

        try {
            checkAge(age);
        } catch (InvalidAgeException e) {
            System.out.println("Exception: " + e.getMessage());
        }

        sc.close();
    }
}
```

Output:

```
Output

Enter your age: 32
You are eligible to vote.
```

2. **Create a Driving License System that throws a custom exception if the user's age is below 18. If the age is valid, display that the user is eligible for a driving license**

Code:

```
import java.util.Scanner;

class InvalidAgeException extends Exception {
    InvalidAgeException(String message) {
        super(message);
    }
}

public class Main {

    static void checkAge(int age) throws InvalidAgeException {
        if (age < 18) {
            throw new InvalidAgeException("Age must be 18 or above.");
        } else {
            System.out.println("You are eligible for a driving license.");
        }
    }

    public static void main(String[] args) {
```

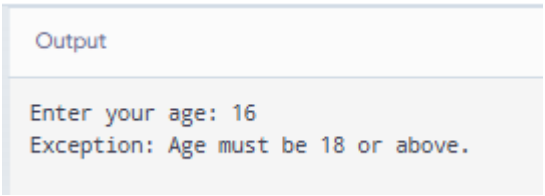
```
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter your age: ");
        int age = sc.nextInt();

        try {
            checkAge(age);
        } catch (InvalidAgeException e) {
            System.out.println("Exception: " + e.getMessage());
        }

        sc.close();
    }
}
```

Output:



Conclusion

Both programs demonstrate the concept of custom exception handling in Java. A custom exception class is created by extending the Exception class. The Voting System checks whether the user is eligible to vote, while the Driving License System checks whether the user is eligible for a driving license. If the user's age is below 18, the throw keyword is used to generate a custom exception. The try-catch block handles the exception and displays an appropriate message. The throws keyword is used to declare that a method may generate an exception. These programs show how custom exceptions can be used to handle invalid input in a simple and controlled manner. Overall, they provide a basic understanding of exception handling and custom exceptions in Java.